Web Development Assignment

1. Difference between Frontend, Backend and Full-Stack Development

Frontend: It is the part of a website that users can see and interact with. It is mainly developed using HTML, CSS and JavaScript. Example: buttons, menus and product pages of Amazon.

Backend: It works behind the website and handles server-side logic, data and requests. Example: login and order processing in Amazon.

Full-Stack: A developer who works on both frontend and backend is called a full-stack developer.

2. Client-Server Model

```
Client (Browser)
   |
 Request
   ↓
 Server
   |
 Response
   ↓
Client (Browser)
```

The client sends a request to the server. The server processes it and sends a response back to the client.

3. How a Browser Requests and Displays a Web Page

When a user enters a website URL, the browser sends a request to the web server. The server processes the request and sends the required HTML, CSS and JavaScript files back. The browser then processes these files and displays the webpage to the user.

Flow:
Browser → Request → Web Server → Response → Browser

4. Tools Required for Web Development

Some common tools required are:

- **VS Code:** Used for writing and editing code.
- **Web Browser:** Used to run and test webpages.
- **HTML:** Creates the structure of a webpage.
- **CSS:** Used for styling and layout.
- **JavaScript:** Adds functionality and interaction.
- **Git:** Used to manage and track code changes.

5. What is a Web Server?

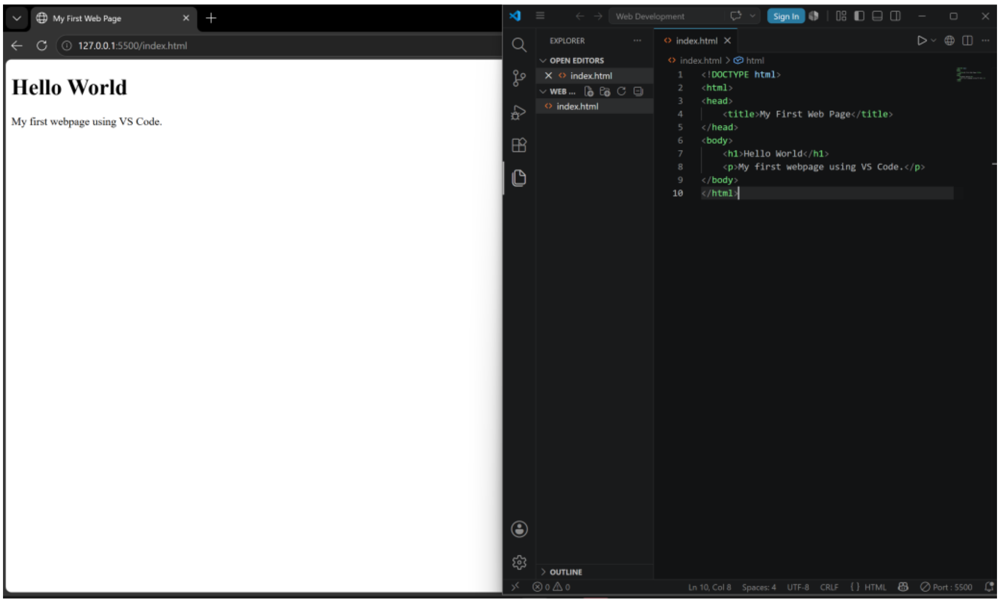
A web server is a system that receives requests from a web browser and sends webpages or data in response. It helps users access websites over the internet.

Examples: Apache, Nginx, Microsoft IIS and LiteSpeed.

6. Roles of Developers and Database Administrator

- **Frontend Developer:** Creates the visible part of a website, such as buttons, menus and pages.
- **Backend Developer:** Handles server-side logic, requests, APIs and application processing.
- **Database Administrator (DBA):** Manages databases, data security, backups and database performance.

7. Install and Configure VS Code



8. Static and Dynamic Websites

Static Website: A static website shows fixed content and usually does not change according to the user.
Example: A simple personal portfolio website.

Dynamic Website: A dynamic website can display different content based on user actions or database information.
Example: Swiggy, Amazon or Instagram.

9. Five Web Browsers and Rendering Engines

A rendering engine converts webpage code like HTML and CSS into the webpage we see on the screen.

Browser	Rendering Engine
Google Chrome	Blink
Mozilla Firefox	Gecko
Microsoft Edge	Blink
Safari	WebKit
Opera	Blink

10. Basic Web Architecture

```
     Client
   (Browser)
      |
      ↓
    Server
      |
 ┌─────┴─────┐
 ↓     ↓
API   Database
 |     |
 └─────┬─────┘
      ↓
   Response
      |
      ↓
    Client
```

Explanation: The client sends a request to the server. The server processes the request and can communicate with APIs and the database. The required data is then sent back to the client.